```c
// Write a program to implement strlen() function.
#include <stdio.h>
int main (){
    int l=0;
    int n=100;
    char str[1000];
    fgets(str,1000,stdin);
    for (int i=0;str[i]!='\0';i++){
        l++;
    }
    printf("the length is %d",--l);
return 0;
}
```

```
~/Documents/dslab  gcc assignment3_Q1.c
~/Documents/dslab  ./a.out
udbhav
the length is 6
~/Documents/dslab  ./a.out
qwerty asdfg
the length is 12
~/Documents/dslab  ./a.out

the length is 0
~/Documents/dslab
```

```
// Write a program to implement strcpy() function.
#include <stdio.h>
int main (){
    char str1[1000];
    char str2[1000];
    fgets(str1,1000,stdin);
    for (int j=0;str1[j]!='\0';j++){
        if (str1[j]=='\n'){
        str1[j]='\0';
        break;
        }}
    int j=0;
    for (j=0;str1[j]!='\0';j++){
        str2[j]=str1[j];
        }
    str2[j]='\0';
    printf("copied one : %s\n",str2);
return 0;
}
```

```
~/Documents/dslab  nvim assignment3_Q2.c
~/Documents/dslab  gcc assignment3_Q2.c
~/Documents/dslab  ./a.out
ud
copied one : ud
~/Documents/dslab  ./a.out
ud bh
copied one : ud bh
~/Documents/dslab  ./a.out
ud bh av
copied one : ud bh av
```

```c
//Write a program to implement strcat() function.
#include <stdio.h>
int main (){
    char str1[1000];
    printf("enter string 1\n");
    fgets(str1,1000,stdin);
    for (int i=0;str1[i]!='\0';i++){
        if (str1[i]=='\n'){
            str1[i]='\0';
                break;
    }}
    printf("enter the string 2\n");
    char str2[1000];
    fgets(str2,1000,stdin);
     for (int i=0;str2[i]!='\0';i++){
        if (str2[i]=='\n'){
            str2[i]='\0';
                break;
    }}
    char nstr[2000];
    int j=0;
    for (int i=0;str1[i]!='\0';i++){
        nstr[j++]=str1[i];
    }
    for (int i=0;str2[i]!='\0';i++){
        nstr[j++]=str2[i];
    }
    nstr[j] = '\0';
    printf("\n%s\n",nstr);
return 0;
}
```

```
~/Documents/dslab  gcc assignment3_Q3.c
~/Documents/dslab  ./a.out
enter string 1
udbhav
enter the string 2
dubey

udbhavdubey
~/Documents/dslab  ./a.out
enter string 1
qwerty
enter the string 2
 asdfgh

qwerty asdfgh
~/Documents/dslab  ./a.out
enter string 1
zxcv gfd
enter the string 2
asdwqe

zxcv gfdasdwqe
```

```
//Write a program to implement strcmp() function.
#include <stdio.h>
int main (){
 char str1[1000];
    printf("enter string 1\n");
    fgets(str1,1000,stdin);
    for (int i=0;str1[i]!='\0';i++){
        if (str1[i]=='\n'){
            str1[i]='\0';
                break;
    }}
    printf("enter the string 2\n");
    char str2[1000];
    fgets(str2,1000,stdin);
     for (int i=0;str2[i]!='\0';i++){
        if (str2[i]=='\n'){
            str2[i]='\0';
                break;
    }}
    int i=0,result=0;
    while(str1[i]!='\0'&&str2[i]!='\0'){
        if (str1[i]!=str2[i]){
            result=str1[i]-str2[i];
            break;
        }
        i++;
    }
    if (result==0){
        result=str1[i]-str2[i];
    }
    if (result==0){printf("strings are equal\n");}
    else if (result<0){printf("string1 is smaller\n");}
    else printf("string 1 is greater\n");
return 0;
}
```

```
~/Documents/dslab  gcc assignment3_Q4.c
~/Documents/dslab  ./a.out
enter string 1
udbhav
enter the string 2
dubey
string 1 is greater
~/Documents/dslab  ./a.out
enter string 1
asdf
enter the string 2
xzc
string1 is smaller
~/Documents/dslab  gcc assignment3_Q4.c
~/Documents/dslab  ./a.out
enter string 1
uijn
enter the string 2
uijm
string 1 is greater
```

```c
//Write a program to demonstrate limitations of Two-Dimensional Array of Characters.
#include <stdio.h>

int main() {
    // 2D array: 3 strings, each can hold max 10 chars (including '\0')
    char names[3][10] = {"Udbhav", "Dubey", "VeryLongNameThatGetsCut"};

    printf("Stored strings:\n");
    for (int i = 0; i < 3; i++) {
        printf("%s\n", names[i]);
    }

    return 0;
}
```

```
~/Documents/dslab  gcc assignment3_Q5.c
                 In function '    ':
                     warning: initializer-string for array of '    ' is too long (24 chars into 10 available)
   6 |     char names[3][10] = {"Udbhav", "Dubey", "VeryLongNameThatGetsCut"};
     |                                             ^~~~~~~~~~~~~~~~~~~~~~~~
                     note: (near initialization for '        ')
~/Documents/dslab  gcc assignment3_Q5.c
```

nvim

```c
//Write a program to demonstrate an array of Pointers to Strings.
#include <stdio.h>
int main() {
    const char *fruits[] = {
        "Apple",
        "Banana",
        "Mango",
        "Watermelon",
        "Strawberry"
    };

    int n = sizeof(fruits) / sizeof(fruits[0]);

    printf("Array of pointers to strings:\n\n");
    for (int i = 0; i < n; i++) {
        printf("fruits[%d] -> %s\n", i, fruits[i]);
    }

    return 0;
}
```

```
~/Documents/dslab  gcc assignment3_Q6.c
~/Documents/dslab  ./a.out
Array of pointers to strings:

fruits[0] -> Apple
fruits[1] -> Banana
fruits[2] -> Mango
fruits[3] -> Watermelon
fruits[4] -> Strawberry
```